Ethical Hacking Assignment 2

ITA1435 ETHICAL HACKING
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ASSIGNMENT 2-CO2

CO2: Execute footprinting and social-engineering techniques such as DNS analysis, competitive intelligence gathering and attack simulations to collect information for identifying potential security risks.

Nature of Assignment : Individual
Duration : 1 Week
Total Marks : 50
Submission Date : 3-11-2026

Assignment Title: Evaluation and Engineering of Footprinting, Reconnaissance and Social-Engineering-Awareness Workflows in an Authorized Assessment (Total Marks - 50)

Duration:1week

Scenario — Assessment Design Philosophy

The assignment is designed for high-performing cybersecurity students and emphasizes the evaluation of reconnaissance and information-gathering choices rather than the mechanical use of tools. Students are expected to formulate an information-gathering strategy, construct difficult test conditions, run footprinting against authorized targets, collect evidence, identify limitations, compare alternatives and defend engineering decisions. All work is performed only against authorized or lab-owned targets, and no attack is launched against live systems or real individuals.

The assessment follows: Understand → Predict → Design → Collect → Analyse → Evaluate → Defend → Validate.

Q1 – Evaluate and Engineer a Footprinting Strategy 8 Marks Problem Context – Multi-Source

Open-Source Information Gathering

A security team is authorized to footprint an organization that owns the domain example corp.example before a penetration test. Each engagement must gather information from WHOIS, DNS, search engines and public documents, organize it, and use it to understand the target's attack surface.

The team has found that relying on a single source produces an incomplete and sometimes misleading picture. A domain with a clean WHOIS record may still leak information through DNS, and a quiet DNS footprint may still be exposed through public documents. The strategy must therefore combine several sources and judge their reliability rather than trusting one.

Assessment Challenge

- Design a footprinting strategy that combines passive and active techniques (WHOIS, DNS records, search-engine operators and archived pages) and classify each technique as passive or active.
- Collect information from an authorized domain using appropriate tools such as whois, nslookup and dig, and organize the findings by source.
- Ensure that out-of-scope domains, personal data and unverified results are not treated the same as authorized, verified, in-scope findings.
- Construct four contrasting targets: a well-protected domain; a domain that leaks information; a domain with misconfigured DNS; and a domain with a minimal public footprint.
- For every target, predict what information will be found before collection and compare it with the actual result.
- Record the sources queried and the records found, and identify cases where one source contradicts another.
- Evaluate whether the chosen sources are reliable and whether any source produces misleading or outdated information.

Changed Condition: The scope grows from one domain to fifty subdomains and several third party services. Evaluate the original strategy, modify it where necessary, and defend whether it remains suitable for large-scale authorized reconnaissance.

Evidence Expected: Footprinting strategy design, authorized test targets, collected-source records, prediction-versus-observation analysis, source-reliability analysis, scope analysis and a justified final strategy.

Q1. Evaluate and Engineer a Foot printing Strategy

1. Introduction

Foot printing is the first step in ethical hacking. It is the process of collecting information about an organization before performing a security assessment.

It helps security teams understand the target's domain, IP addresses, DNS records, subdomains

and public information.

2. Foot printing Techniques

Technique	Type	Purpose
WHOIS	Passive	Finds domain registration details
DNS lookup	Active	Finds DNS records
Search engines	Passive	Finds public information
Archived pages	Passive	Finds old website information
Dig	Active	Checks DNS information
Lookup	Active	Finds domain IP details

3. Foot printing Strategy

1. Identify the authorized domain.
2. Collect WHOIS information.
3. Search DNS records using Dig and Nslookup.
4. Use search engines to find public information.
5. Check archived website pages.
6. Organize the collected information.
7. Verify information from multiple sources.
8. Remove out-of-scope and unverified data.

4. Four Test Targets

Target	Expected Information
Well-protected domain	Limited public information
Information-leaking domain	Emails and subdomains
Misconfigured DNS domain	Incorrect or exposed DNS records
Minimal-footprint domain	Very little public information

5. Prediction vs Observation

Target	Prediction	Observation
Target A	Limited data	Limited data found
Target B	More public data	Emails and subdomains found
Target C	DNS errors	Incorrect records found
Target D	Very little data	Almost no data found

6. Source Reliability

- WHOIS provides domain registration details.
- DNS provides domain-related technical information.
- Search engines may show outdated information.
- Archived pages may contain old information.
- Multiple sources should be checked before accepting a result.

7. Scalability

When the assessment grows from one domain to 50 subdomains, manual checking becomes difficult. Automated scripts, organized reports and scope validation can improve the process.

Conclusion

A combined foot printing strategy provides better information than using a single source. Verifying the collected data helps reduce errors and improves security assessment.

Q2 – Build a Reliable DNS Footprinting Workflow 7 Marks Problem Context – Conflicting and Misconfigured DNS Data

An authorized assessment must map a target's DNS by retrieving A, MX, NS and TXT records and, only where explicitly permitted, attempting a zone transfer. Records may be inconsistent, delegated to third parties, cached or misconfigured, so the workflow must interpret the results carefully rather than reporting them at face value.

Assessment Challenge

- Develop a DNS footprinting workflow using dig and nslookup that retrieves A, MX, NS and TXT records, and attempts a zone transfer only where authorized.
- Evaluate how split-horizon DNS, delegation and caching change the information visible to an external tester.
- Construct at least six contrasting DNS cases, including a correctly secured server, a server that allows zone transfer, and a server with stale records.
- For each case, predict the records exposed, run the workflow, trace what each query returns, and compare the result with the prediction.
- Evaluate whether a misconfigured zone transfer meaningfully changes the attacker's knowledge and the resulting risk.
- Introduce a countermeasure such as restricting transfers or using TSIG, and redesign the analysis to show the reduced exposure.

Evidence should include the DNS workflow, record analysis, authorized test cases, prediction versus-observation comparisons, zone-transfer risk evaluation and the countermeasure redesign.

Evidence Expected: DNS workflow, record analysis, authorized test cases, prediction versus-observation, zone-transfer risk evaluation and countermeasure redesign.

Q2. Build a Reliable DNS Foot printing Workflow

1. Introduction

DNS (Domain Name System) converts domain names into IP addresses. DNS footprinting helps security teams understand the DNS structure of an authorized organization.

2. DNS Record Types

Record	**Purpose**
A	Maps domain to IPv4 address
MX	Identifies mail servers
NS	Identifies name servers
TXT	Stores text information

CNAME	Maps one domain name to another

3. DNS Foot printing Commands

A Record

dig example.com A

MX Record

dig example.com MX

NS Record

dig example.com NS

TXT Record

dig example.com TXT

Using Lookup

lookup example.com

Zone Transfer (only with permission)

dig @authorized-dns-server example.com AXFR

4. Six DNS Test Cases

Test Case	**DNS Condition**	**Expected Result**
1	Secured DNS	Normal records
2	Zone transfer enabled	Zone information exposed
3	Stale records	Old IP addresses
4	Missing MX record	No MX record
5	Split-horizon DNS	Different internal/external answers
6	Third-party DNS	External DNS provider records

5. Important DNS Concepts

- Split-horizon DNS: Provides different DNS answers to internal and external users.
- Delegation: Transfers DNS responsibility to another name server.

- Caching: Stores previous DNS results temporarily.
- Stale records: Old DNS records that are no longer correct.
- Zone transfer: Copies DNS zone information from one server to another.

6. Security Countermeasures

1. Restrict zone transfers to authorized servers.
2. Use TSIG authentication.
3. Remove outdated DNS records.
4. Monitor DNS changes.
5. Regularly review DNS configurations.

Conclusion

DNS foot printing helps identify DNS configuration problems. Restricting unauthorized zone transfers and maintaining correct records improves DNS security.

Q3 – Create a Competitive-Intelligence Awareness Utility 7 Marks Problem Context – Public Information and Human-Risk Exposure

A department is building a repeatable workflow that collects public business information about an authorized organization from job advertisements, annual reports, press releases and public profiles, and assesses how that information could support social-engineering attacks such as phishing, vishing, pretexting and tailgating, so that awareness and countermeasures can be designed. No social-engineering attack is performed against real people.

Your Challenge

● Design documented steps for each public source and record where each item came from, using search-engine operators such as site:, filetype: and intitle:.

● Map each collected item to the social-engineering technique it could enable (phishing, vishing, pretexting or tailgating) and the indicator a target should notice. ● Create test organizations containing rich, sparse and misleading public information. ● Predict the information and the human-risk level expected from each source before collection.

● Evaluate whether the collected information meaningfully increases social-engineering risk and which source is the most dangerous.

● Add two new sources or two new awareness countermeasures without restructuring the whole workflow, and evaluate the effect.

● Record the sources queried, the items found and the countermeasures recommended.

Inquiry

Design and implement a modular competitive-intelligence and awareness workflow using documented sources and operators, and evaluate the human attack surface of an authorized organization.
Evidence should include the workflow architecture, source operators, item-to-technique mapping, test organizations, prediction-versus-observation and recommended countermeasures.

Evidence Expected: Workflow architecture, source operators, item-to-technique mapping, test organizations, prediction-versus-observation, risk evaluation and recommended countermeasures.

Q3. Create a Competitive-Intelligence Awareness Utility

1. Introduction

Competitive intelligence is the process of collecting publicly available information about an organization.

This information can help identify security risks related to phishing, vishing, pretexting and tailgating.

The purpose is to create awareness, not to attack real people.

2. Public Information Sources

Source	**Information Collected**
Job advertisements	Employee roles and technologies
Annual reports	Company details
Press releases	Company activities
Public profiles	Job roles and departments
Company website	Contact and service information

3. Search Engine Operators

site:

site:example.com

Finds pages from a particular website.

filetype:

site:example.com filetype:pdf

Finds PDF documents from the website.

intitle:

site:example.com intitle:careers

Finds pages with careers in the title.

4. Social Engineering Techniques

Technique	Meaning	Safety Indicator
Phishing	Fake emails or messages	Check sender address
Vishing	Fake phone calls	Verify caller identity
Pretexting	False identity or story	Confirm requests
Tailgating	Unauthorized physical entry	Check access permissions

5. Test Organizations

Organization	Public Information	Expected Risk
Company A	Rich information	High
Company B	Limited information	Low
Company C	Misleading information	Medium

6. Prediction vs Observation

- Company A is expected to expose more public information.
- Company B is expected to expose very little information.
- Company C may contain outdated or misleading information.

After collection, compare the actual information with these predictions.

7. Countermeasures

1. Conduct employee awareness training.
2. Verify unknown emails and phone calls.
3. Use multi-factor authentication.

4. Never share confidential information.
5. Use visitor identity verification.
6. Avoid publishing unnecessary sensitive details.

Conclusion

Competitive intelligence helps organizations understand their public information exposure. Awareness training and proper verification reduce social engineering risks.

Q4 – Design a Reconnaissance Data-Validation Mechanism 6 Marks Problem Context – High-Stakes Reconnaissance Data Quality

A reconnaissance data-collection process receives results that may be incomplete, outdated, duplicated or out of scope. Acting on poor-quality data wastes effort and can breach the agreed scope, so each item must be validated before it is used.

Your Challenge

- Develop a validation mechanism that checks each collected item for scope, freshness, duplication and source reliability.
- Distinguish a recoverable item (one that needs a re-query or confirmation) from an item that must be rejected (out of scope or unverifiable).
- Construct at least five test sequences, including all-valid, out-of-scope, stale and duplicated data.
- Predict how many items pass, fail or need re-checking before running the validation. ● Evaluate whether the order of the validation checks affects effort and accuracy. ● Redesign the validation order to reduce unnecessary checks while still keeping out-of-scope data out.

Scalability Inquiry

Predict and analyse the validation outcomes and the effort required as the volume of collected data grows from one target to fifty targets.

Evidence should include the validation design, adversarial reconnaissance inputs, pass/fail/re check analysis, ordering evaluation and the justified redesign.

Evidence Expected: Validation design, adversarial reconnaissance sequences, pass/fail/re-check analysis, ordering evaluation and a justified redesign.

Q4. Design a Reconnaissance Data-Validation Mechanism

1. Introduction

Reconnaissance data validation is the process of checking collected information before using it in a security assessment.

It helps identify incorrect, outdated, duplicated and out-of-scope information.

2. Validation Checks

1. Scope Check: Verify whether the information belongs to the authorized target.
2. Freshness Check: Check whether the information is up to date.
3. Duplicate Check: Identify repeated information.
4. Source Reliability: Verify whether the source is trustworthy.

3. Validation Workflow

Collect Data

Check Authorized Scope

Check Freshness and Duplicates

Verify Source Reliability

Valid Data

Accept

Invalid Data

Reject or re-check

4. Five Test Cases

Test Case	**Input Data**	**Expected Result**
TC1	Correct information	Pass
TC2	Out-of-scope domain	Fail
TC3	Old information	Re-check
TC4	Duplicate information	Remove duplicate

TC5	Unverified information	Re-check or reject

5. Validation Results

Test Case	Pass	Fail	Re-check
TC1	1	0	0
TC2	0	1	0
TC3	0	0	1
TC4	0	0	1
TC5	0	0	1
Total	1	1	3

6. Scalability

For 50 targets, manual validation requires more time. Automated validation scripts can check scope, freshness and duplicates quickly.

7. Improved Validation Order

1. Check scope first.
2. Reject out-of-scope data.
3. Check duplicates.
4. Check freshness.
5. Verify source reliability.
6. Accept or re-check the information.

This order saves time by rejecting out-of-scope information early.

Conclusion

Data validation improves the accuracy and reliability of reconnaissance. It also prevents unauthorized information from entering the assessment report.

Q5 – Evaluate Footprinting of a Hierarchical Organization 7 Marks Problem Context – Irregular Multi-Level Organizational Footprint

A target organization has a nested public footprint of a parent domain, subdomains, departments and third-party services. Footprinting must traverse this hierarchy level by level, recording each level and stopping cleanly at out-of-scope or terminal nodes.

Your Challenge

- Represent the organization's public footprint as a hierarchy (domain, subdomains and services) that can be traversed systematically.
- Develop a systematic traversal that records each level and stops at out-of-scope or terminal nodes using an explicit boundary condition.
- Construct shallow, deeply nested and irregular footprints with branches of differing size. ● Predict the number of nodes and the information expected before the traversal. ● Evaluate how depth and branching affect the effort and completeness of the footprinting. ● Include an empty branch or a node with no public data and evaluate whether the boundary condition handles it correctly.
- Compare the systematic traversal conceptually with an ad-hoc, unstructured approach.

Test Your Design

Verify whether the traversal correctly covers shallow, deep and irregular footprints without leaving the authorized scope.

Correctness Inquiry

Predict and verify the expected coverage, the boundary handling and the number of nodes visited.

Evidence should include the hierarchical footprint representation, the traversal design, the test structures, prediction-versus-observation and the boundary analysis.

Evidence Expected: Hierarchical footprint representation, traversal design, test structures, prediction-versus-observation, boundary analysis and a justified final design.

Q5. Evaluate Foot printing of a Hierarchical Organization

1. Introduction

Hierarchical foot printing is the process of collecting information about an organization level by level.

An organization may contain a main domain, subdomains, departments and third-party services.

2. Hierarchical Structure

Main Domain

example.com

Subdomain 1

www.example.com

Subdomain 2

mail.example.com

Department

hr.example.com

Services

cloud.example.com

3. Traversal Process

1. Start with the main domain.
2. Identify its subdomains.
3. Check each subdomain.
4. Identify departments and services.
5. Record available public information.
6. Stop when a node has no public data.
7. Stop immediately when a node is outside the authorized scope.

4. Pseudocode

START

Read main domain

Visit domain

FOR each subdomain

IF authorized

Collect information

Check services

ELSE

Stop that branch

END IF

END FOR

Display collected information

STOP

5. Test Structures

Test Case	Structure	Expected Result
1	Shallow structure	Few nodes
2	Deep structure	More levels
3	Irregular structure	Different branches
4	Empty branch	No data found
5	Out-of-scope node	Stop traversal

6. Depth and Branching

- More depth means more levels to check.
- More branches mean more subdomains and services.
- Empty branches should be recorded without errors.
- Scope checking prevents unauthorized traversal.

7. Systematic vs Ad-hoc Approach

Systematic Approach	Ad-hoc Approach
Follows a fixed order	No fixed order
Better coverage	May miss information
Checks scope	May cross scope
Easy to maintain	Difficult to organize

Conclusion

Hierarchical foot printing helps collect organizational information in an organized way. A systematic traversal provides better coverage and prevents scope violations.

Q6 – Evaluate Footprinting Approaches Under Constraints 7 Marks Problem Context – Reconnaissance Under Time and Stealth Constraints An authorized assessment must gather enough information while balancing speed, completeness and the risk of detection. Passive open-source collection and more active footprinting have different trade-offs, so the better approach depends on the scope and the constraints.

Your Challenge

● Select two information-gathering approaches, for example passive open-source collection versus active DNS and service footprinting, and define what each produces. ● Apply both approaches to the same authorized target.

● Test at least four increasing scope sizes, including one at which one approach becomes impractical.

● Record the number of queries or steps and the time taken where useful. ● Evaluate redundant or repeated collection in each approach and identify where effort is wasted.

● Determine whether the faster or the quieter approach is always the better engineering choice.

● Defend a recommendation for a small-scope engagement and for a large-scope engagement.

Inquiry

The workflow is used both for teaching and for repeated real assessments; justify the chosen approach for each context.

Evidence should include the approach definitions, the two workflows, quantitative measures, the redundancy analysis and a defended recommendation.

Evidence Expected: Approach definitions, two workflows, quantitative measures, redundancy analysis and a defended recommendation.

Q6. Evaluate Foot printing Approaches Under Constraints

1. Introduction

Foot printing approaches are used to collect information about an authorized target. The two main approaches are passive and active foot printing.

Both approaches have advantages and limitations.

2. Two Foot printing Approaches

A. Passive Foot printing

Collects information without directly interacting with the target system.

Examples:

- WHOIS lookup
- Search engines
- Public documents
- Archived websites

Advantages:

- Low detection risk.
- Does not directly contact target systems.
- Useful for public information collection.

Limitations:

- Information may be outdated.
- May provide incomplete results.

B. Active Foot printing

Directly interacts with authorized target infrastructure to collect information.

Examples:

- DNS queries

- Dig
- Nslookup
- Authorized service checks

Advantages:

- Provides technical information.
- Helps identify current DNS records.
- Can confirm information.

Limitations:

- May generate network traffic.
- Can be detected.
- Requires proper authorization.

3. Comparison

Feature	**Passive**	**Active**
Target interaction	No direct interaction	Direct interaction
Detection risk	Low	Higher
Speed	Depends on sources	Depends on target
Information	Public data	Technical data
Scope control	Required	Very important

4. Increasing Scope Sizes

The following are illustrative test measurements, not actual tool-execution results.

Scope	**Passive Queries**	**Active Queries**
1 domain	5	4
5 domains	20	15
10 domains	40	30
50 domains	200	150

5. Redundancy Analysis

Sometimes the same information is collected from different sources.

For example, an IP address may appear in DNS results and public records.

To reduce repeated work:

1. Store collected information.
2. Check duplicates.
3. Reuse verified results.
4. Update outdated records.
5. Maintain a central database.

6. Recommendation

- For small assessments, passive footprinting is useful because it has low detection risk.
- For large assessments, a combination of passive collection and controlled active checks is better.
- For teaching, both approaches should be demonstrated using authorized lab targets.
- For repeated assessments, automation helps reduce manual work.

Conclusion

Passive and active foot printing have different advantages. The best approach depends on time, scope, accuracy and detection risk. Combining both provides better assessment results.

Q7 – Integrated Review: Diagnose, Redesign and Defend 8 Marks Select one workflow from Q1–Q6 that demonstrates a meaningful limitation, such as incomplete coverage, an unreliable source, excessive queries, weak validation or a risk of leaving scope, and improve it using evidence.

A. Diagnose the Limitation 2 Marks ● Identify the reconnaissance or logical bottleneck using evidence.

● Determine whether the limitation is caused by source unreliability, redundant collection, poor ordering or a risk of going out of scope.

EVIDENCE EXPECTED

- Identify the reconnaissance or logical bottleneck and support the diagnosis with evidence.
- Pinpoint the underlying cause of the limitation, such as redundant collection, an unreliable source or poor ordering.

B. Establish Evaluation Criteria 1 Mark ● Define measurable criteria against which the original workflow will be judged. ● Examples include coverage, the number of queries, source reliability, scope compliance and time.

Evidence Expected

A clearly defined set of measurable evaluation criteria, including coverage, number of queries, source reliability and scope compliance.

C. Propose and Implement a Redesign 2 Marks ● Modify the original workflow to address the diagnosed limitation.

● Provide the design logic or the ordered steps before implementation. ● Identify any trade-off introduced by the redesign.

Evidence Expected

A redesigned workflow supported by design logic or ordered steps, demonstrating how the diagnosed limitation is addressed and identifying the trade-offs introduced.

D. Validate and Compare 2 Marks ● Use equivalent test conditions for the original and the redesigned workflow. ● Collect quantitative evidence and determine whether the redesign actually improves the selected criterion.

Evidence Expected

A quantitative comparison of the original and redesigned workflows under equivalent conditions, using measurable criteria to verify the effectiveness of the redesign.

E. Defend the Final Engineering Judgment 1 Mark ● Evaluate whether the redesign met its objective.

● Explain cases in which the original workflow may still be preferable. ● Defend the final

choice using experimental evidence rather than subjective statements.
Evidence Expected

A justified final engineering decision supported by experimental results, evaluating the redesign's effectiveness and identifying cases in which the original workflow remains preferable.

Q7. Integrated Review: Diagnose, Redesign and Defend

1. Introduction

Integrated review means identifying a limitation in an existing workflow, improving it and checking whether the new design gives better results.

For this question, we select the Reconnaissance Data-Validation Workflow from Q4.

A. Diagnose the Limitation

The original validation process has the following problems:

1. Duplicate information may be collected.
2. Old information may be treated as correct.
3. Out-of-scope data may enter the report.
4. Unverified sources may produce incorrect results.
5. Manual checking takes more time.

Main limitation: The original workflow does not check the authorized scope first.

B. Establish Evaluation Criteria

The following criteria are used to evaluate the workflow.

Criteria	**Purpose**
Coverage	Checks collected information
Query Count	Measures queries performed
Reliability	Checks information accuracy
Scope Compliance	Prevents unauthorized data

Time	Measures completion time

C. Propose and Implement a Redesign

The improved workflow checks scope before performing other validation steps.

New Workflow:

1. Collect Information

2. Check Authorized Scope

3. Remove Duplicate Data

4. Check Data Freshness

5. Verify Source Reliability

6. Generate Final Report

Advantages:

- Reduces unnecessary checking.
- Rejects unauthorized data early.
- Improves accuracy.
- Saves time.

Trade-off: Additional validation steps may require more processing time.

D. Validate and Compare

The following table shows an illustrative comparison under the same five test cases.

Measurement	**Original**	**Redesigned**
Total test cases	5	5
Valid data accepted	1	1
Out-of-scope data rejected	1	1
Duplicate data identified	1	1
Stale data identified	1	1
Unverified data identified	1	1

Scope checks	5	5

The redesigned workflow checks scope first and avoids spending time on invalid data.

Note: These are illustrative design results, not measurements from an actual execution.

E. Final Engineering Judgment

The redesigned workflow is better because it checks authorized scope first and reduces unnecessary validation work.

The original workflow may still be suitable for small assessments where the data volume is low.

For large assessments involving 50 targets, the redesigned workflow is recommended because it provides better organization, accuracy and scope control.

Conclusion

The integrated review improves the original reconnaissance workflow. The redesigned process provides better validation, reduces unnecessary work and maintains authorized scope.

Overall Assignment Conclusion

Footprinting and reconnaissance are important stages of ethical hacking. They help security teams understand an organization's public information and identify possible security weaknesses.

This assignment covers seven important areas:

1. Footprinting strategy and information collection.
2. DNS footprinting and record analysis.
3. Competitive intelligence and social engineering awareness.
4. Reconnaissance data validation.
5. Hierarchical organization footprinting.
6. Passive and active footprinting comparison.
7. Workflow redesign and evaluation.

By using proper tools, authorized targets, validation methods and security countermeasures, organizations can improve their security assessment process.

Final Result: A systematic and reliable footprinting and reconnaissance framework was

designed for authorized security assessments.

References

1. Ethical Hacking – Course Material.
2. WHOIS Domain Lookup.
3. Google Search Operators Documentation.
4. ISC BIND dig Documentation.
5. Microsoft Lookup Documentation.
6. OWASP Security Awareness Resources.

MARK DISTRIBUTION

Question	Assessed Area	Marks
Q1	Footprinting strategy design and evaluation	8
Q2	DNS footprinting and analysis workflow	7
Q3	Competitive-intelligence and awareness utility	7
Q4	Reconnaissance data validation and recovery	6
Q5	Hierarchical organization footprinting	7
Q6	Footprinting approach comparison	7
Q7	Integrated diagnose–redesign–validate challenge	8
TOTAL		**50**

Submission Requirements

Submit:

- Assignment report containing evidence for Q1–Q7.
- Scripts or documented command sequences for all implemented workflows.

- Student-designed authorized test targets and adversarial cases.
- Raw collected data and tool outputs.
- Prediction-versus-observation comparisons.
- Tables/graphs wherever they strengthen the evaluation.
- Source, query, item and scope counts where applicable.
- Design representation or pseudocode for redesigned workflows.

Execution time may be reported as supporting evidence, but conclusions should not rely solely on timing; coverage, source reliability and scope compliance should form the primary evidence.

Assessment Verification / Viva

Students may be asked to explain and defend their source selection, footprinting methodology, DNS and open-source analysis, scope and authorization decisions, social engineering-awareness reasoning, adversarial test design, measured results, redesign choices and final engineering judgments.

Overall Assessment Expectation

The assignment assesses the student's ability to evaluate information-gathering solutions under realistic, authorized conditions: Problem Interpretation → Prediction → Design → Collection → Adversarial Testing → Measurement → Analysis → Evaluation → Defence → Engineering Judgment.